

Smart Metro Booking System

Project Planning Phase

Date	
Team ID	
Project Name	Smart Metro Booking System
Maximum Marks	5 Marks

1. Project Planning Phase

1. Purpose

This document defines the project planning approach for digitizing metro ticket booking using ServiceNow. It translates the solution requirements and technology stack into a structured plan — covering scope, work breakdown, timeline, roles and risk management — so that the implementation proceeds in a controlled, milestone-driven manner.

2. Project Scope & Deliverables

The planning phase covers the full delivery of the ServiceNow-based metro ticketing solution, from catalog configuration through go-live. Key deliverables include:

- A configured Service Catalog item with Source, Destination, Passenger Type and Number of Tickets variables.
- Automated fare calculation logic including concessions for Student / Senior Citizen categories.
- QR code generation and validation integrated with the booking and payment flow.
- Station Details Mapping covering routes, distances and fare slabs.
- Passenger-facing UI pages across Mobile App, Portal and WhatsApp channels.
- A tested and deployed production-ready solution following UAT sign-off.

3. Work Breakdown Structure

The project is broken down into four major work streams — Requirements & Design, Development, Integration, and Testing & Deployment — each further divided into concrete work packages aligned to the project milestones.

Work Breakdown Structure — Metro Digital Ticketing

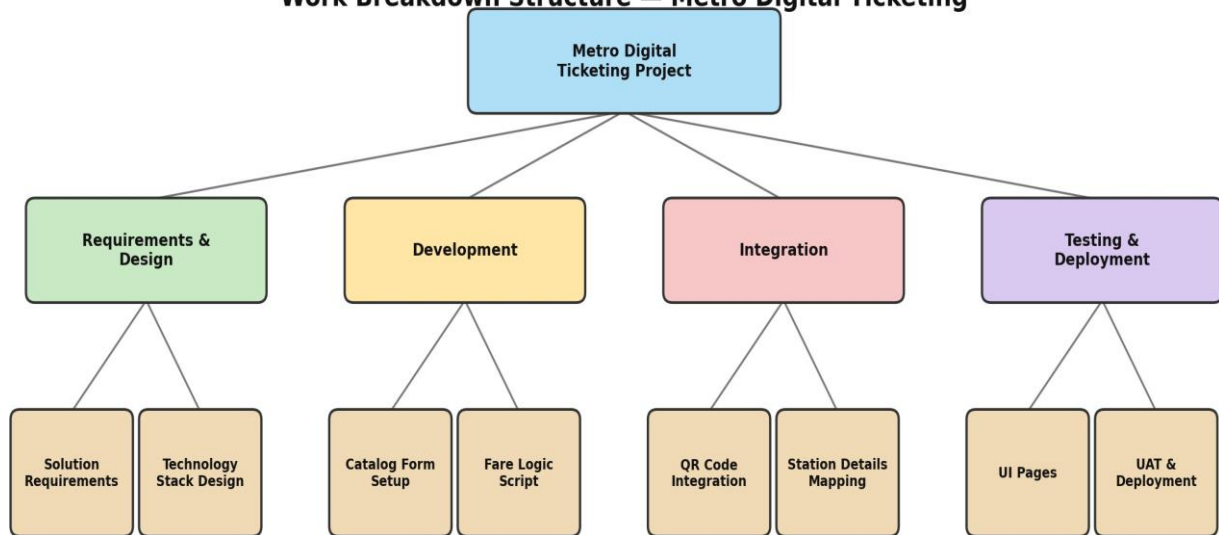


Figure 1 — Work Breakdown Structure for Metro Digital Ticketing

4. Project Timeline & Milestones

The implementation follows six sequential milestones. The timeline below reflects planned durations in project weeks, with Fare Logic Script and Station Details Mapping running partly in parallel with Catalog Form Setup to compress the overall schedule.

Project Timeline — Metro Digital Ticketing Implementation

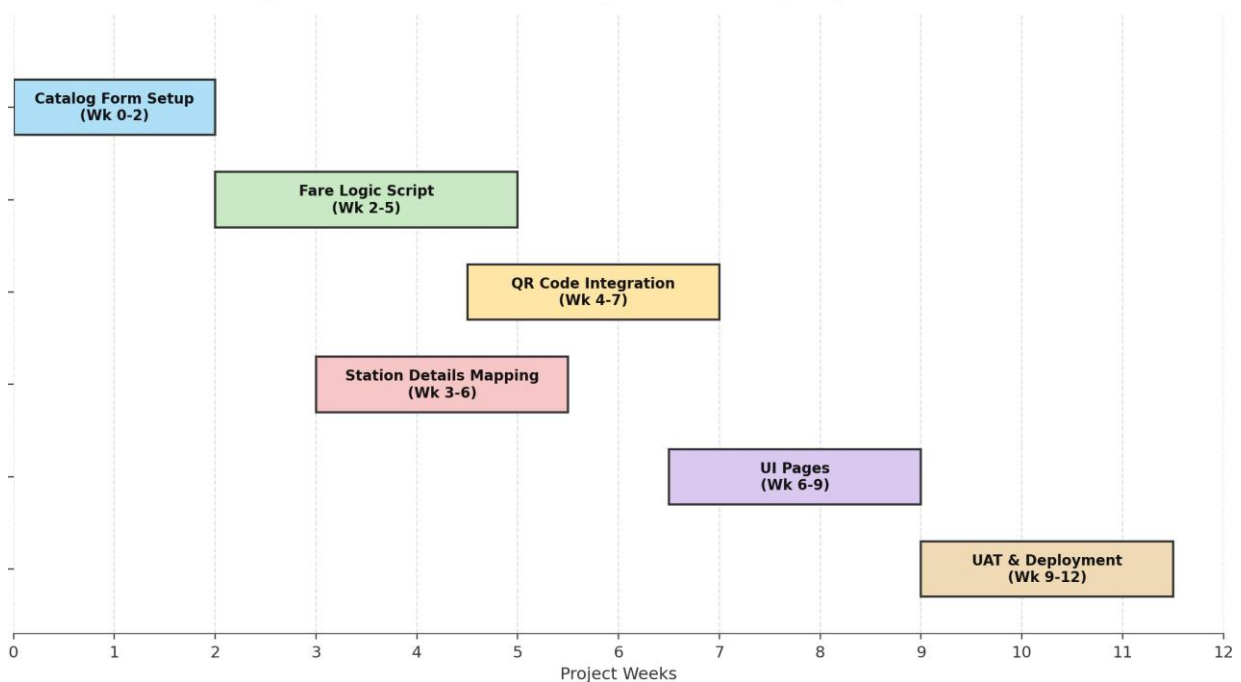


Figure 2 — Indicative Project Timeline (in weeks)

4.1 Milestone Details

Milestone	Description	Owner
Catalog Form Setup	Configure Service Catalog item with booking variables and validations.	IT Admin / ServiceNow Dev
Fare Logic Script	Build fare calculation business rules and concession logic.	ServiceNow Developer
QR Code Integration	Integrate QR generation API into the post-payment workflow.	Integration Developer
Station Details Mapping	Set up station master data, routes and fare slabs.	Metro Operations Team
UI Pages	Build passenger-facing booking and ticket display screens.	UI/UX Developer
UAT & Deployment	User acceptance testing and production rollout.	QA Team / Project Manager

5. Roles & Responsibilities (RACI)

The RACI matrix below clarifies who is Responsible, Accountable, Consulted and Informed for each major milestone.

Milestone	Passengers	Station Mgr	Ops Team	IT Admin
Catalog Form Setup	I	I	C	R/A
Fare Logic Script	I	I	C	R/A
QR Code Integration	I	I	I	R/A
Station Details Mapping	I	C	R/A	C
UI Pages	C	I	I	R/A
UAT & Deployment	C	C	A	R

R = Responsible, A = Accountable, C = Consulted, I = Informed.

6. Risk Management Plan

Risk	Impact	Mitigation
Delay in payment gateway API access/credentials	High	Initiate vendor onboarding in parallel with Catalog Form Setup.
Incomplete or inaccurate station master data	Medium	Validate station and fare data with Operations Team before Fare Logic Script begins.
QR code duplication or security gaps	High	Use signed/encrypted QR tokens and test tamper scenarios during UAT.
Low passenger adoption of digital channels	Medium	Run pilot at select stations with in-person support before full rollout.
AFC gate hardware compatibility issues	Medium	Conduct early hardware integration testing during QR Code Integration milestone.

7. Assumptions & Constraints

- ServiceNow instance and required licenses (Catalog, Flow Designer, IntegrationHub) are provisioned before the planning phase starts.
- Payment gateway and QR generation vendors are finalized and API access is available.
- Station Master Data is owned and maintained by the Metro Operations Team.
- UAT will be conducted with representative passengers, Station Managers and IT Admins.
- Go-live is contingent on successful UAT sign-off from all stakeholder groups.